Mohammad Saneian

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About me

I'm a PhD graduate of Computer Science at Northeastern University. I'm an experienced coder and problem solver with 10+ years of experience in algorithm design and implementation. I'm passionate about technology, algorithms, AI, and software engineering.

Education

- **Northeastern University** — Boston, USA
  PhD in computer science — Sep 2022 - Dec 2025
  - **Advisor:** Soheil Behnezhad and Mahsa Derakhshan
  - **Thesis title:** Matching algorithms under uncertainty
- **Sharif University of Technology** — Tehran, Iran
  B.Sc. in Computer Engineering — Sep 2017 - Feb 2022
  - **Thesis title:** Simple streaming edge coloring algorithms

Experience

- **Algorithmic problem solving**
  2015-Present
  - One problem I tackled during my PhD was the streaming edge coloring problem. In this problem, we have $O(n \log n)$ memory to edge color all edges of the graph such that no two incident edges receive the same color. Our novel idea reduces the problem to a case where the graph is bipartite, which we call the two parts online and offline vertices. We use a random number for each offline vertex and color an edge using this number. We show that this approach colors most of the edges with high probability, and we then use a new color palette to color the rest of the edges.
  - Have solved **700+** problems in the Codeforces website and achieved maximum rating of **Grandmaster**.
- **Rahnema College** — Tehran, Iran
  Data Science Intern — Jan 2020 - Feb 2020
  - Gained some knowledge around Machine learning and Deep learning.
  - Became familiar with Big data technologies like Hadoop and Spark.
  - Did a team project for detecting fraudulent comments on Sanjagh website based on their extracted features.
- **EPFL (Ecole polytechnique federale de Lausanne)** — Lausanne, Switzerland
  Research Intern — Jul 2020 - Mar 2021
  - In this internship, our project was attacking trajectory prediction models. We studied the impact of localization errors on their predictions, which is essential for real-world applications.

Honors and Awards

- **Silver Medalist** in the 29$^{\text{th}}$ International Olympiad of Informatics (IOI) — 2017
- **Onsite contest finalist** at Snackdown19, ranked **22nd** among **25,000** teams — 2019
- **Silver Medalist** in the 11$^{\text{th}}$ Asia-Pacific Informatics Olympiad (APIO) — 2017
- **Gold Medalist** in the 26$^{\text{th}}$ Iranian National Olympiad in Informatics (INOI) — 2016
  Gold medals are awarded to 8 people selected after a year of competition among over 10000 students.
- **Bronze medalist** in the 25$^{\text{th}}$ Iranian National Olympiad in Informatics (INOI) — 2015
- **Gold medalist** in the ACM ICPC regional contest — 2018
- **Winner** of ACM-style contests: FCPC-2019, Amirkabir 2019 — 2019

Publications

- **Half-Approximating Maximum Dicut in the Streaming Setting**
  A. Azarmehr, S. Behnezhad, S. Ferrante, M. Saneian
  Published in STOC 2026
- **A Simple Analysis of Ranking in General Graphs**
  M. Derakhshan, M. Roghani, M. Saneian, T. Yu
  Published in SOSA 2026
- **Improved Approximation for Ranking on General Graphs**
  M. Derakhshan, M. Roghani, M. Saneian, T. Yu
  Published in SODA 2026
- **Query Efficient Weighted Stochastic Matching**
  M. Derakhshan, M. Saneian
  Published in ICALP 2025
- **Streaming Edge Coloring with Asymptotically Optimal Colors**
  S. Behnezhad, M. Saneian
  Published in ICALP 2024
- **Query Complexity of Stochastic Minimum Vertex Cover**
  M.Derakhshan, M.Saneian
  Published in ITCS 2025
- **Are socially-aware trajectory prediction models really socially-aware?**
  S. Saadatnejad, M. Bahari, P. Khorsandi, M. Saneian, M. Moosavi-Dezfooli, A.Alahi
  Published in Journal of Transportation Research Part C , 2022
- **Simple Streaming Algorithms for Edge Coloring**
  M. Asari, M. Saneian, H. Zarrabi-Zadeh
  Published in ESA 2022

Skills

- **Programming**: Python (Proficient), C++ (Proficient), Java (Proficient), R, Sql, Bash
- **Machine Learning Frameworks:**: PyTorch, Keras, NumPy, Scikit-Learn, Pandas
- **Web Development & Database**: HTML, CSS, JavaScript, Docker, PostgreSQL, Spark
- **Miscellaneous**: Git, Linux, Windows, LaTeX, Microsoft Office (Word, Excel, Powerpoint)

PhD courses

- **CS7880**: Special Topics in TCS: Algorithms for Big Data
- **CS7600**: Intensive computer systems
- **CS7870**: A Modern Toolkit for Algorithms and Analysis
- **CS7870**: Seminar in Theoretical Computer Science
- **CS5610**: Web development
- **CS7200**: Statistical methods in machine learning